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Metric Spaces

1.1 Convergence of Sequences

Definition 1.1.1 ► Convergence

Let (X, d) be a metric space and $x_1, \dots, x_n, \dots$ a sequence of points in X . Then the sequence is said to **converge to** $x \in X$ if given any $\varepsilon > 0$ there exists an integer n_0 such that for all $n \geq n_0$, $d(x, x_n) < \varepsilon$. This is denoted by $x_n \rightarrow x$.

The sequence $y_1, y_2, \dots, y_n, \dots$ of points in (X, d) is said to be **convergent** if there exists a point $y \in X$ such that $y_n \rightarrow y$.

Theorem 1.1.2

Let (X, d) be a metric space. $A \subseteq X$ is a subset, $p \in A$. Then p is a limit point of A , if and only if there exists a sequence $\{a_n\}$ composed with distinct points in A converging to p .

Proof. If p is a limit point of A . Then $B_d(p, 1) \cap (A \setminus \{p\}) \neq \emptyset$, we choose a point a_1 there. If we have points $a_1, a_2, \dots, a_n$ such that $a_i \in B_d\left(p, \frac{1}{i}\right) \cap B_d(p, m_i)$, where

$$m_i := \min \{d(p, a_j)\}_{1 \leq j \leq i}$$

Since the sets $\left(B_d\left(p, \frac{1}{n+1}\right) \cap B_d(p, m_n)\right) \cap (A \setminus \{p\})$ is nonempty, we can take a point in it. We inductively give a sequence $\{a_n\}$ composed with distinct points in A .

If $\{a_n\}$ is such sequence. Let U be any open set containing p . Since $\{B_d(x, \varepsilon) : x \in X, \varepsilon > 0\}$ is a basis for (X, d) . We have from Proposition ?? that there exists a ball B such that $p \in B \subseteq U$. Then it is easy to verify that there exists $B_d(p, \varepsilon)$ such that $p \in B_d(p, \varepsilon) \subseteq B$. Since the sequence $\{a_n\}$ in A converges to p , there exists some (in fact, infinitely many) $a_n \in B_d(p, \varepsilon)$ that is not equal to p . i.e, $p \in B_d(p, \varepsilon) \cap (A \setminus \{p\}) \subseteq U \cap (A \setminus \{p\}) \neq \emptyset$. $\square$

Proposition 1.1.3

Let (X, d) be a metric space. A subset A of X is closed in (X, d) if and only if every convergent sequence of points in A converges to a point in A . (In other words, A is closed in (X, d) if and only if $a_n \rightarrow x$, where $x \in X$ and $a_n \in A$ for all n , implies $x \in A$.)

1.2 Completeness

Definition 1.2.1 ▶ Cauchy sequence

A sequence $x_1, x_2, \dots, x_n, \dots$ of points in a metric space (X, d) is said to be a **Cauchy sequence** if given any real number $\varepsilon > 0$, there exists a positive integer n_0 , such that for all integers $m \geq n_0$ and $n \geq n_0$, $d(x_m, x_n) < \varepsilon$.

Proposition 1.2.2

Let (X, d) be a metric space and $x_1, x_2, \dots, x_n, \dots$ a sequence of points in (X, d) . If there exists a point $a \in X$, such that the sequence converges to a , that is, $x_n \rightarrow a$, then the sequence is a Cauchy sequence.

Definition 1.2.3

A metric space (X, d) is said to be **complete** if every Cauchy sequence in (X, d) converges to a point in (X, d) .

Proposition 1.2.4

Let (X, d) be a metric space, Y a subset of X , and d_1 the metric induced on Y by d .

- (i) If (X, d) is a complete metric space and Y is a closed subspace of (X, d) , then (Y, d_1) is a complete metric space.
- (ii) If (Y, d_1) is a complete metric space, then Y is a closed subspace of (X, d) .

1.3 The Metric Topology

Theorem 1.3.1 ► standard bounded metric

Let X be a metric space with metric d . Define $\bar{d} : X \times X \rightarrow \mathbb{R}$ by the equation

$$\bar{d}(x, y) = \min\{d(x, y), 1\}.$$

Then $\bar{d}$ is a metric that induces the same topology as d .

Remark.

The metric $\bar{d}$ is called the **standard bounded metric** corresponding to d .

Proof. In any metric space, the collection of ε -balls with $\varepsilon < 1$ forms a basis for the metric topology. d and $\bar{d}$ induce the same topology on X since there ε -balls with $\varepsilon < 1$ under these two metrics are the same collection. $\square$

Theorem 1.3.2

Let $\bar{d}(a, b) = \min\{|a - b|, 1\}$ be the standard bounded metric on $\mathbb{R}$. If $\mathbf{x}$ and $\mathbf{y}$ are two points of $\mathbb{R}^\omega$, define

$$D(\mathbf{x}, \mathbf{y}) = \sup \left\{ \frac{\bar{d}(x_i, y_i)}{i} \right\}.$$

Then D is a metric that induces the product topology on $\mathbb{R}^\omega$.

Definition 1.3.3

Given an index set J , and given points $\mathbf{x} = (x_\alpha)_{\alpha \in J}$ and $\mathbf{y} = (y_\alpha)_{\alpha \in J}$ of $\mathbb{R}^J$, let us define a metric $\bar{\rho}$ on $\mathbb{R}^J$ by the equation

$$\bar{\rho}(\mathbf{x}, \mathbf{y}) = \sup\{\bar{d}(x_\alpha, y_\alpha) \mid \alpha \in J\},$$

where $\bar{d}$ is the standard bounded metric on $\mathbb{R}$. It is easy to check that $\bar{\rho}$ is indeed a metric; it is called the **uniform metric** on $\mathbb{R}^J$, and the topology it induces is called the **uniform topology**.

Theorem 1.3.4

The uniform topology on $\mathbb{R}^J$ is finer than the product topology and coarser than the box topology; these three topologies are all different if J is infinite.